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Syn2real: Forgery Classification via Unsupervised Domain Adaptation

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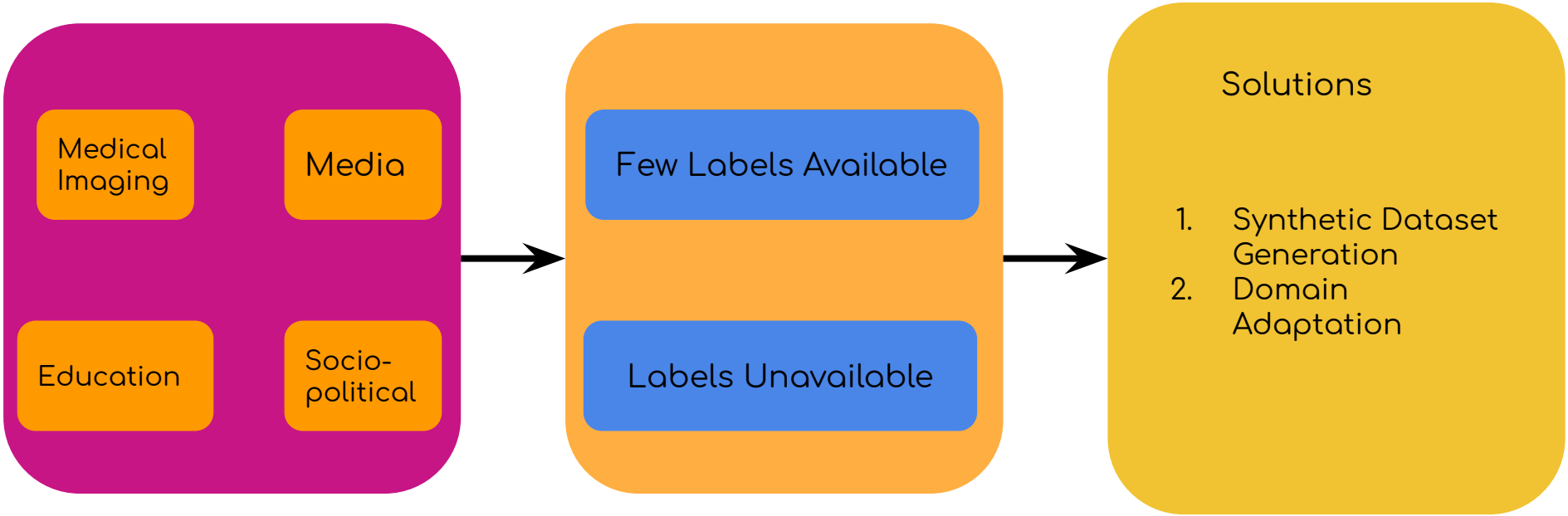
What is Copy-Move Forgery?

- A section of image is copied and pasted within the same image.
- Many post-image processing applications are applied to easily blend in the manipulated region.

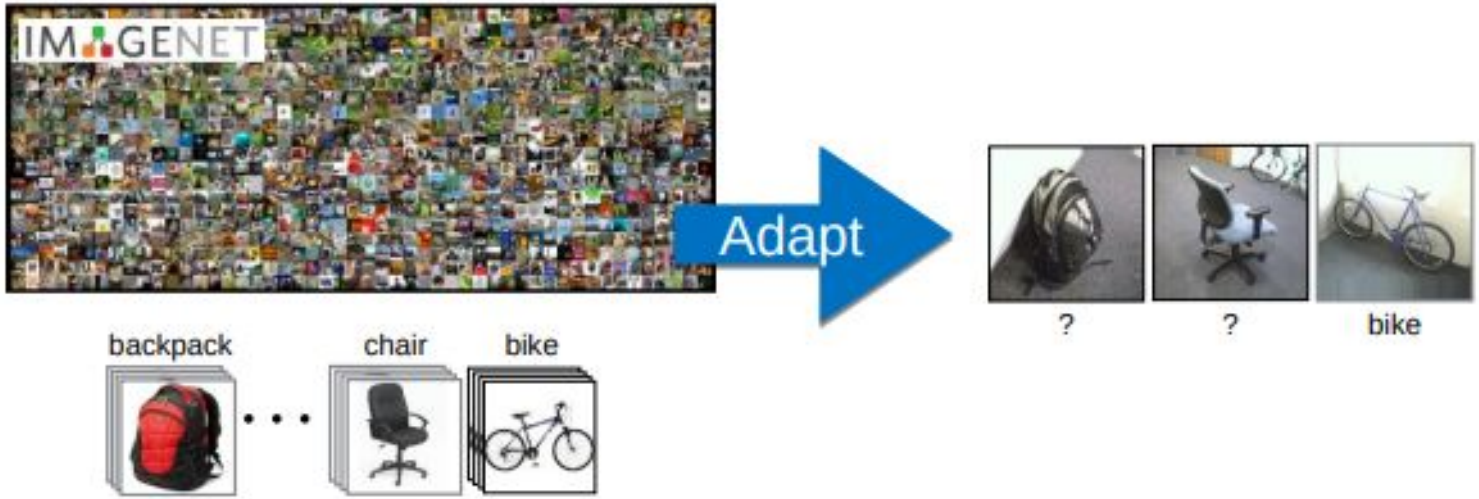
Why study Copy-Move forgery specifically?

- Exponential rise compared to other types of forgery
- Hard to detect
- As the source and target image remains the same, the photometric characteristics of image remain mostly invariable.

Problem Statement



Unsupervised Domain Adaptation



Source Domain $\sim P_S(X, Y)$ $\neq$ Target Domain $\sim P_T(Z, H)$

lots of **labeled** data

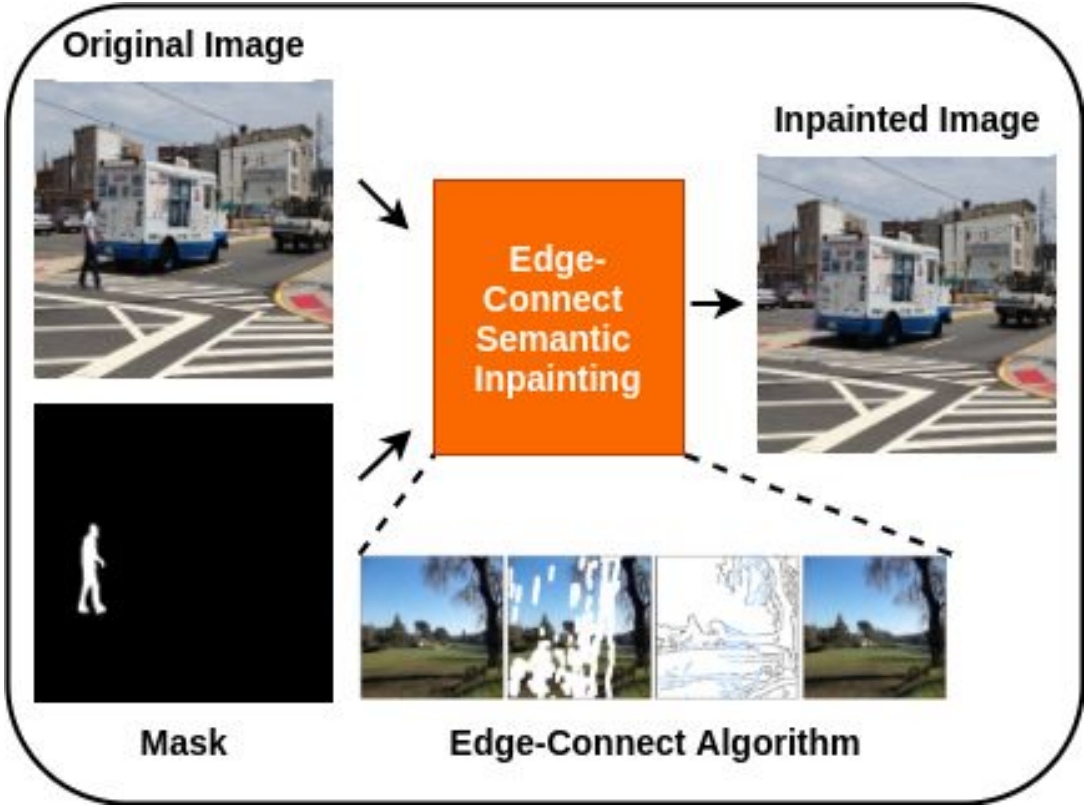
unlabeled or limited labels

$$D_S = \{(\mathbf{x}_i, y_i), \forall i \in \{1, \ldots, N\}\}$$

$$D_T = \{(\mathbf{z}_j, ?), \forall j \in \{1, \ldots, M\}\}$$

Credits: J. Hoffman et al.

Dataset Generation: Semantic Inpainting



Dataset Generation: CMF Approach

Datasets Overview

Source Domain

Target Domain

Authentic

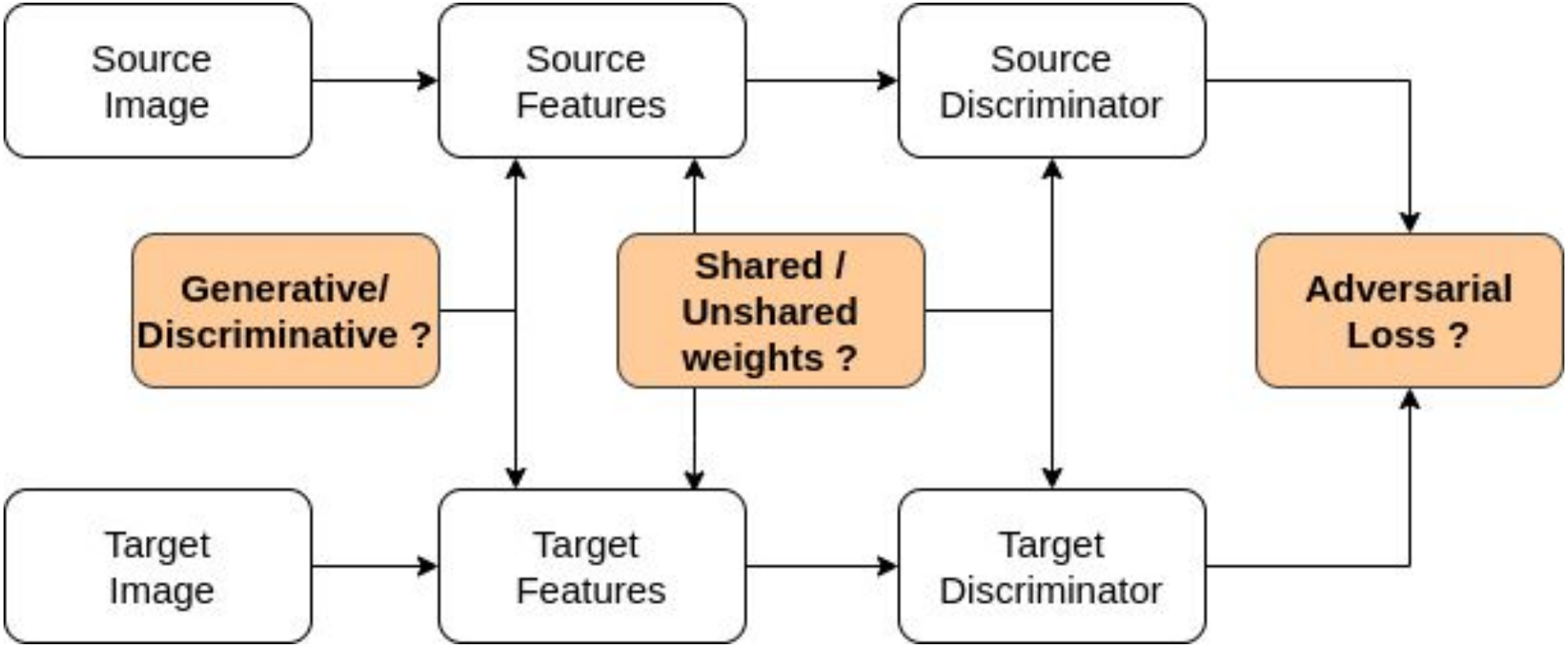
Forged

COCO Synthetic

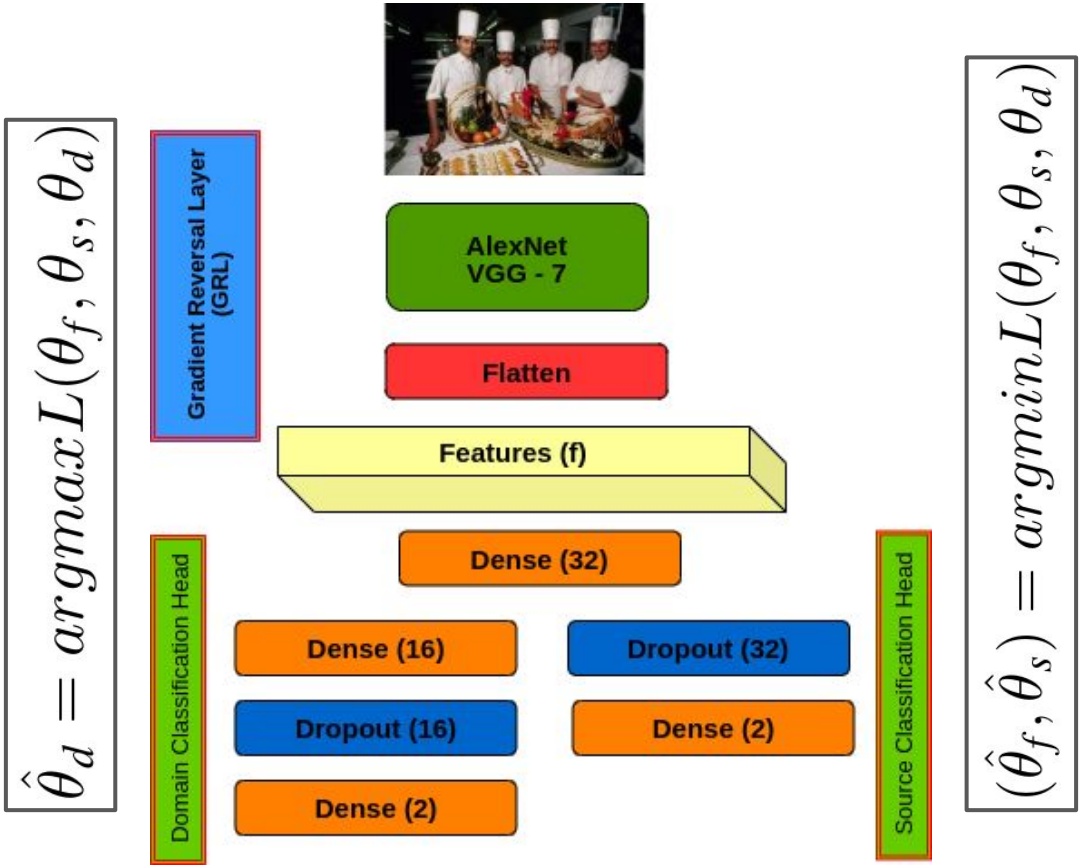
CASIA

CoMoFoD

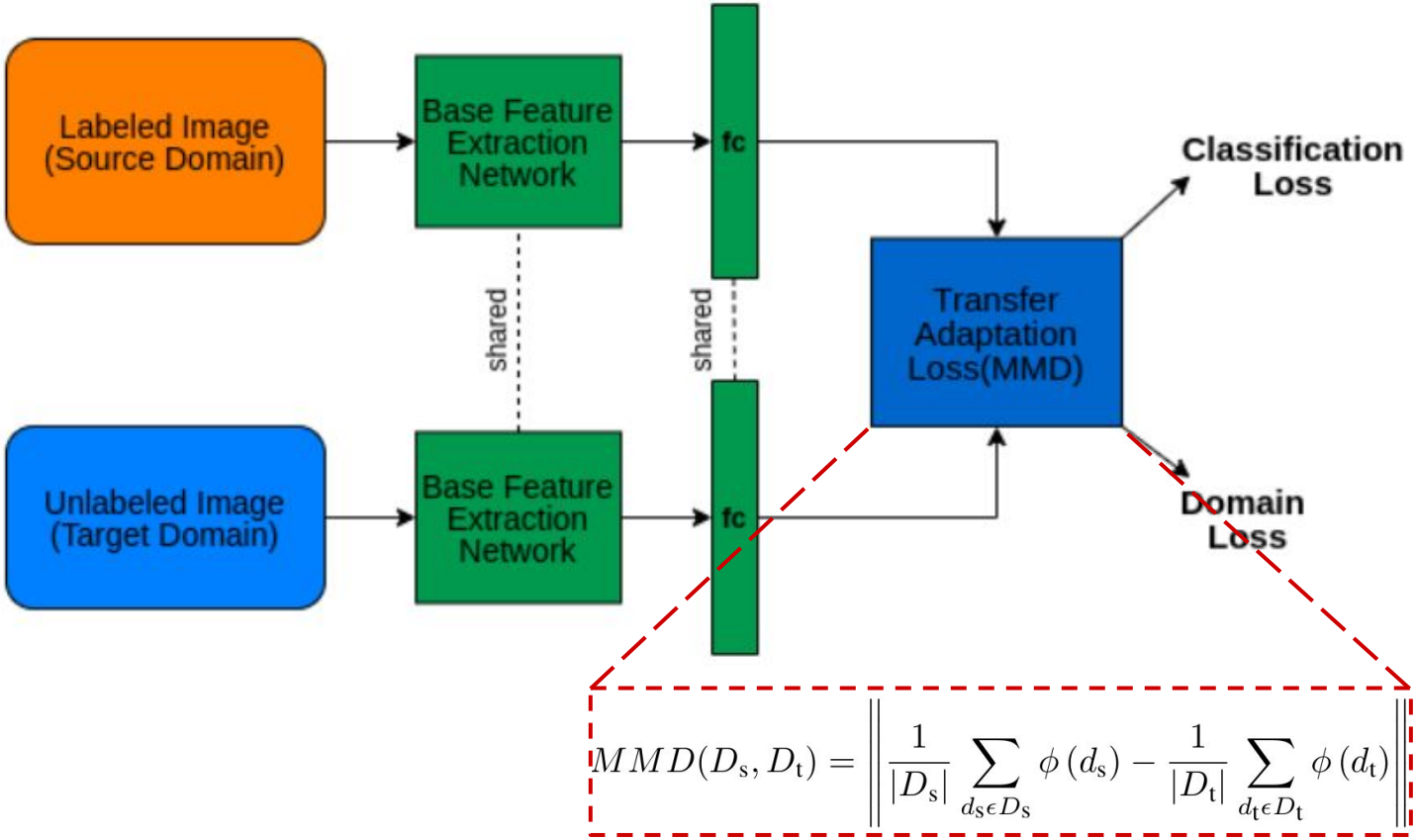
Generalized Domain Adaptation categorization



Domain Adversarial Neural Network



Deep Domain Confusion



Losses:

- Ds: Source Data Distribution
- Dt: Target Data Distribution
- Df: Features distribution
- Ls: Source Classification Loss
- Ld: Domain Classification Loss

- DANN Loss

$$L(\theta_f, \theta_s, \theta_d) = \sum_{x=1\ldots N} L_{\mathrm{s}}(D_{\mathrm{s}}(D_{\mathrm{f}}), y_{\mathrm{i}}) + (\Delta) \sum_{x=1\ldots N} L_{\mathrm{d}}(D_{\mathrm{t}}(D_{\mathrm{f}}), y_{\mathrm{i}})$$

- DDC Loss

$$L_{\mathrm{total}} = L_{classify}(x, y) + \alpha MMD^2(D_{\mathrm{s}}, D_{\mathrm{t}})$$

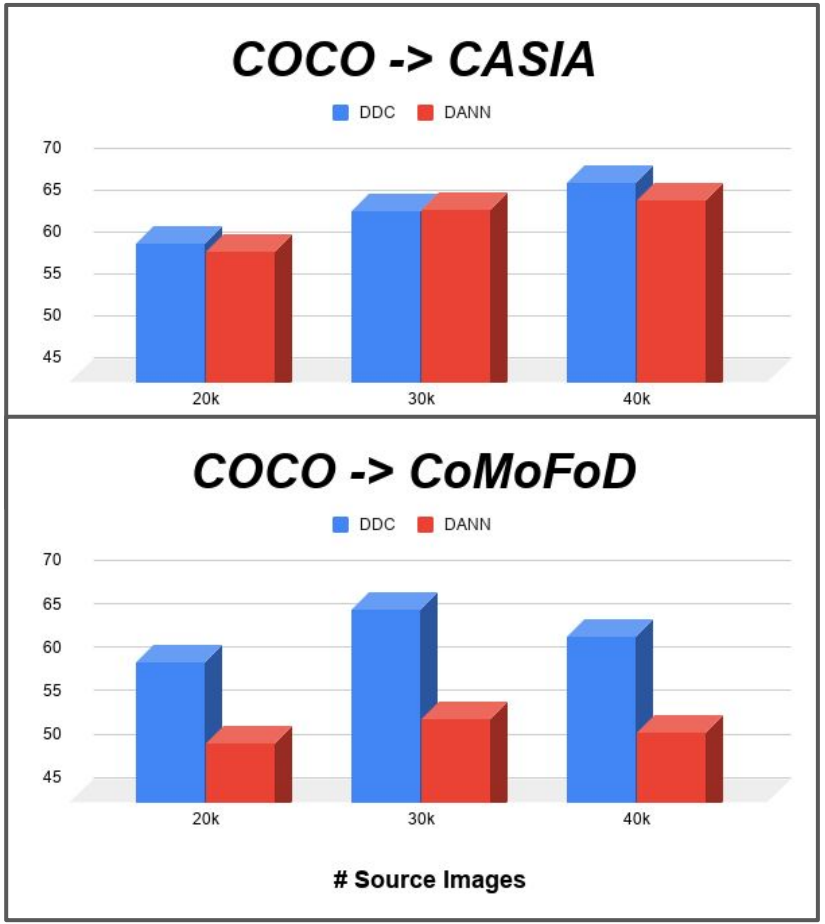
Experiments: Datasets Characteristics

		CASIA	CoMoFoD
1	Resolution	240x160 - 900x600	512x512
2	# pristine/tampered	1701/3274	200/200
3	Image Format	JPG, TIFF	PNG
3	Post-image processing	Translation, Rotation, Scaling, Affine Transformation	Translation, Rotation, Scaling, Affine Transformation

Results:

- *COCO -> CASIA:*
  - Train/Test: 4k/1k images.
  - More images -> Able to optimize huge number of parameters.
  - DANN+VGG outperforms others.

- *COCO -> CoMoFoD:*
  - Train/Test: 200/200 images.
  - Less images -> Can't optimize huge number of parameters.
  - DDC+AlexNet outperforms others.



Results:

Architecture	CASIA			CoMoFoD		
	Precision	Recall	F-1	Precision	Recall	F-1
BusterNet [17]	**78.22**	73.89	75.98	57.34	49.39	49.26
DDC MMD AlexNet	68.02	97.73	80.18	**65.62**	**97.86**	**78.99**
DDC MMD VGG	57.27	95.02	70.13	65.39	91.24	76.18
DANN AlexNet	66.19	98.47	79.16	51.05	83.5	63.34
DANN VGG	68.26	**99.84**	**80.34**	49.65	72.04	58.75

- Compare to BusterNet where they have used **1 lakh** images for supervised training, we used **40k** images to achieve better accuracy.
- Our approach using **Domain Adaptation** improves the previously reported baseline.

Thanks!

Any questions?

You can find me at

- @github.com/AKASH2907 **[Code]**
- mailto:kmr2907akash@gmail.com